Normal Distribution

Section 1: The Standard Normal Distribution $Z \sim N(0,1)$

1 Find $P(Z < 1.645)$ for the standard normal distribution $Z \sim N(0,1)$.

a. 0.9500

b. 0.9750

c. 0.9000

d. 0.9900

2 Find $P(Z > -1.282)$ for $Z \sim N(0,1)$.

a. 0.9000

b. 0.1000

c. 0.8000

d. 0.8500

3 Find $P(-1.96 < Z < 1.96)$ for $Z \sim N(0,1)$.

a. 0.9500

b. 0.9750

c. 0.9000

d. 0.9900

Section 2: Standardization & General Normal Distribution

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4 The random variable $X \sim N(50, 16)$. Find $P(X < 56)$.

a. 0.9332

b. 0.8413

c. 0.9772

d. 0.8849

5 Heights of adult females are normally distributed with mean 165 cm and standard deviation 6 cm. What percentage of females are taller than 177 cm?

a. 2.28%

b. 5.00%

c. 4.56%

d. 1.14%

6 If $Y \sim N(100, 225)$, find $P(85 < Y < 115)$.

a. 0.6826

b. 0.9544

c. 0.5000

d. 0.7500

Section 3: Inverse Normal & Finding μ and σ

7 The random variable $X \sim N(\mu, 25)$ has $P(X < 62) = 0.8413$. Find the mean μ.

a. 57

b. 55

c. 52

d. 60

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8 For a normal variable X, $P(X < 40) = 0.05$ and $P(X < 70) = 0.95$. Find μ and σ.

a. $\mu = 55, \sigma = 9.12$

b. $\mu = 55, \sigma = 15.0$

c. $\mu = 50, \sigma = 10.0$

d. $\mu = 55, \sigma = 5.50$

9 Packets of biscuits have nominal weight 200 g. Actual weights are normally distributed with $\mu = 204$ g and $\sigma = 2.5$ g. What proportion of packets are underweight (< 200 g)?

a. 0.0548

b. 0.0228

c. 0.0100

d. 0.0808

Section 4: Normal Approximation to the Binomial Distribution

10 State the two conditions required to approximate a binomial distribution $X \sim B(n, p)$ by a normal distribution.

a. $np > 5$ and $n(1-p) > 5$

b. $n > 30$ and $p = 0.5$

c. $np > 10$ only

d. $n(1-p) > 100$

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11 A fair coin is tossed 100 times. Using the normal approximation with continuity correction, what is the expression for $P(X \geq 55)$?

a. $P(Y > 54.5)$ where $Y \sim N(50, 25)$

b. $P(Y > 55.5)$ where $Y \sim N(50, 25)$

c. $P(Y > 54.5)$ where $Y \sim N(50, 5)$

d. $P(Y > 55.0)$ where $Y \sim N(50, 25)$

12 Compute the approximate probability of getting at least 55 heads in 100 tosses using $Y \sim N(50, 25)$.

a. 0.1841

b. 0.1587

c. 0.2119

d. 0.1250

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